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Lecture 9a: Computer Misuse & Criminal Law (CMA 1990)

Lecture 9b: Cyber Laws in Pakistan (ETO 2002 · Bill 2007 · PECA 2015)

Lecture 10: Software Contracts & Liability

Lecture 11: BCS Code of Conduct & Good Practice

Case Studies: 8 Exam Scenarios with Full Model Answers

EXAM FORMAT NOTE

No direct questions from slides — paper contains ONLY case studies.

Always cite: exact law name, year, offence number, and clause in your answers.

All 4 lectures can appear together in one case study scenario.

What is Computer Misuse?

Accessing or doing something illegally by using a computer.

7 Types of Computer Misuse

1. Hacking

Unauthorized network access to computers/servers without password. Hackers use tools to break into servers.

Example: Using tools to break into a web server

2. Data Misuse

Copying and illegal transfer of personal data, company research, novels, textbooks without permission.

Example: Copying company files to a personal drive

3. Copyright Piracy

Copying music, movies, software and distributing online without copyright holder's permission.

Example: Sharing pirated movies on the internet

4. Email / Chat Abuse

Impersonation, deception in chat rooms, spreading rumors, spam emails advertising legal and illegal products.

Example: Mass spam campaigns, fake identities in chat rooms

5. Identity & Financial

Misuse of stolen/fictional credit card numbers to obtain goods/services; financial frauds; printing counterfeit money.

Example: Using stolen credit cards for online purchases

6. Fraud & Theft

Stealing personal information and money by providing wrong content on the internet.

Example: Displaying expensive pictures to sell cheap items

7. Viruses

Simple programs written to cause nuisance or damage to computers and their files.

Example: Malware that destroys hard drive data

Computer Misuse Act 1990 (CMA) — 3 Criminal Offences

The CMA is an Act of UK Parliament (1990) designed to criminalize unauthorized access to computer systems and deter criminals from computer crime.

Offence	Description	Example	Punishment
1 Unauthorized Access	Accessing computer material without permission	Looking at someone else's files	Up to 6 months prison + up to £5,000 fine
2 Access with Intent	Unauthorized access WITH intent to commit further criminal offences	Hacking into a bank's computer to steal money	Up to 5 years prison + UNLIMITED fine
3 Unauthorized Alteration	Altering, damaging or destroying computer data without permission	Deleting data or destroying data	Up to 10 years prison + UNLIMITED fine

Prevention of Computer Misuse

Prevention Methods

- Data Protection Act
 - Copyright Law
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- Close down abusive chat rooms

What is Cyber Crime?

Activity in which computers or networks are a tool, a target, or a place of criminal activity. Includes fraud, stealing identities, and violating privacy. The first recorded cyber crime was in 1820.

Types of Cyber Crime

Type	Example
Financial crimes	Cheating, credit card frauds, money laundering
Sale of illegal articles	Narcotics, weapons, wildlife
Online gambling	Online gambling, money laundering
Email spoofing	Emails appearing from a different source than actual sender
Forgery	Counterfeit currency, stamps, mark sheets
Cyber Defamation	Publishing or distributing defamatory material online
Cyber Stalking	Victimizing someone online

Pakistan's 3 Key Cyber Laws — Timeline

2002 — ETO — Electronic Transaction Ordinance

Aim: Recognize and facilitate electronic documents, records, information, communications and transactions. Essential prerequisite for e-commerce growth.

- Increased e-commerce and e-transactions
- Legal and safe trading protection for buyers and sellers
- Reduced transaction costs for small and medium enterprises
- Violation of privacy of information is an offence
- Damage to information system (alter, modify, delete, transmit, store) is an offence

Key Rule:

All offences non-bailable, compoundable and cognizable. No court inferior to Court of Sessions shall try any offence.

2007 — Electronic Crimes Bill (ECB)

Aim: Prevention of Electronic Crimes. FIA given EXCLUSIVE powers to investigate and charge all cases.

- Covers 17 types of cyber crimes
- Applies to EVERY PERSON regardless of nationality or citizenship
- Penalties range from 6 months to LIFE imprisonment
- Capital punishment possible for cyber terrorism
- FIA has exclusive investigative and charging authority

Key Rule:

FIA (Federal Investigation Agency) has EXCLUSIVE powers. Capital punishment for cyber terrorism with Rs. 10 million fine.

2015 — PECA — Prevention of Electronic Crimes Act

Aim: Comprehensive cybercrime legislation. Police/FIA/any agency does NOT need a warrant to search, seize, or arrest.

- Unauthorized access: 6 months or Rs. 100,000 fine
- Unauthorized data copying: 6 months or Rs. 100,000 fine
- Cyber terrorism: 14 years and Rs. 5 million fine
- Electronic fraud: 2 years and Rs. 10 million fine
- Identity crime: 3 months or Rs. 50,000 fine
- ISPs, cafes, malls, hotels, airports must hold data records for 3 months (Section 26)
- Government can block/remove access to any website it deems inappropriate (Section 31)

Key Rule:

Section 31: Govt can block any website. Section 26: All internet providers must keep logs for 3 months.

Electronic Crimes Bill 2007 — Full Punishment Table

Offence	Imprisonment	Fine (Rs.)
Criminal Access	3 years	3 Lac
Criminal Data Access	3 years	3 Lac
Data Damage	3 years	3 Lac
System Damage	3 years	3 Lac
Electronic Fraud	7 years	7 Lac
Electronic Forgery	7 years	7 Lac
Misuse of Device	3 years	3 Lac
Unauthorized Access to Code	3 years	3 Lac
Malicious Code	5 years	5 Lac
Defamation	5 years	5 Lac
Cyber Stalking	3 years	3 Lac
Cyber Spamming	6 months	50,000
Spoofing	3 years	3 Lac
Pornography	10 years	—
CYBER TERRORISM	LIFE	10 Million

What is a Contract?

A contract is an agreement between two or more persons that can be enforced in a court of law. All parties must intend to make a contract, be competent (old enough, sound mind), and there must be a consideration — each party receives something and provides something.

5 Types of Software Contracts

1. Fixed Price Contract (Bespoke System) — Developer bears risk

Tailor-made system configured for client's specific needs. 3 parts: (1) Agreement, (2) Standard Terms & Conditions, (3) Schedules / Annexes.

- PROS:
- Predictable budget for client
- Developer motivated to be efficient
- Clear deliverables and acceptance tests

- CONS / WATCH OUT:
- Developer adds risk contingency — making it more expensive
- Penalty clauses are rare in practice
- Disputes arise from unauthorized requirement changes

Key Contract Elements:

- Products & Deliverables — what is to be produced and delivered (source code, manuals, test data, training)
- Rights Ownership — intellectual property, copyright, trademarks
- Confidentiality / Non-disclosure agreement
- Payment Terms — stage payments: 15% signature, 65% development, 25% acceptance, 10% end of warranty
- Payments for Delays & Changes — compensates developer when CLIENT fails to meet obligations
- Penalty Clauses — reduce payment for developer delays (e.g. \$5,000/week up to \$100,000 max)
- Client Obligations — documentation, staff access, machine facilities, accommodation
- Working Standards — which party's procedures apply
- Progress Meetings — signed minutes have CONTRACTUAL SIGNIFICANCE as evidence
- Project Managers — both parties must nominate in writing, with explicit financial authority
- Acceptance Procedure — client provides fixed set of tests and expected results; tests cannot be added later
- Warranty — typically 90 days; errors corrected free of charge
- Inflation — clause to adjust charges in lengthy projects
- Indemnity — each party indemnifies the other for their own IP rights violations
- Termination — developer paid for all work done + staff redeployment compensation
- Arbitration — BCS or IEE appoints arbitrator; Arbitration Act 1996 applies
- Applicable Law — which country's laws govern

2. Contract Hire (Body Shopping) — Client manages staff

Developer agrees to supply the client with a certain number of staff at agreed daily or hourly charge rates. The CLIENT takes responsibility for managing the staff.

- PROS:
- Simple contract
- Flexible staffing levels
- Client controls the work direction

- CONS / WATCH OUT:
- Intellectual property rights must still be addressed
- Quality depends on client's management ability

Key Contract Elements:

- Daily or hourly charge rates specified

- Rate depends on experience and qualifications of staff
- Client takes responsibility for managing the supplied staff
- Intellectual property rights ownership must be addressed

3. Consultancy — Expert gives advice

An expert is called in to assess some aspect of operations or strategy and make proposals for improvement. End product is usually a report.

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| <ul style="list-style-type: none"> ● PROS: ● Usually fixed price ● Simpler contract ● Access to specialist knowledge | <ul style="list-style-type: none"> ● CONS / WATCH OUT: ● Difficult to prove that a report 'failed' ● Client relies on consultant's reputation ● Liability often limited |
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Key Contract Elements:

- Confidentiality — consultant learns a lot about the company and could misuse it
- Terms of Reference — explicitly stated; commonest source of disagreement
- Liability — consultant limits liability for loss; client may require professional liability insurance
- Control over the report — who owns/controls publication

4. Time & Materials (Cost Plus) — Client bears risk

Supplier undertakes development but payment is based on actual costs incurred. Developer is NOT committed to complete. Best for poorly-specified or evolving requirements.

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| <ul style="list-style-type: none"> ● PROS: ● Ideal when requirements are not well specified ● No contingency markup in the price ● No acceptance testing disputes | <ul style="list-style-type: none"> ● CONS / WATCH OUT: ● Client bears ALL risk of cost overruns ● No fixed budget guarantee ● No delay payment provisions |
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Key Contract Elements:

- Labor charged same way as contract hire
- Developer has NO commitment to complete the work
- A maximum payment cap may be set, after which the project is reviewed
- No formal acceptance testing
- No delay payment clauses
- Risk of cost overrun transferred to the client

5. Outsourcing (Facilities Management) — Complex shared risk

Client hands over planning, management and operation of certain functions to a specialist developer. Led by UK Civil Service due to difficulty retaining IT staff.

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| <ul style="list-style-type: none"> ● PROS: ● Specialist expertise for non-core functions ● Client can focus on core business ● May reduce permanent staff costs | <ul style="list-style-type: none"> ● CONS / WATCH OUT: ● Highly complex contracts ● Very difficult to reverse if unsatisfactory ● Requires robust SLA and audit rights |
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Key Contract Elements:

- Service Level Agreement (SLA) — MOST CRITICAL element; often treated as a separate agreement
- How is performance to be monitored and managed?
- What happens if performance is unsatisfactory?
- Which assets are being transferred?

- Staff transfers
- Audit rights — client's right to inspect developer's operations
- Contingency plan and disaster recovery
- Intellectual property rights
- Duration of agreement
- Termination provisions

Fixed Price — Stage Payment Pattern

Stage	% of Total	When Due	Purpose
Contract Signature	15%	On signing	Initial commitment
Development Stages	65%	Various milestones	Ongoing cash flow
Acceptance	25%	On software acceptance	Completion incentive
End of Warranty	10%	After warranty period	Quality assurance

Liability for Defective Software

Standard Practice	Developers include limitation clauses. Most contracts limit liability to purchase price or a fixed figure.
UCTA 1977	Unfair Contract Terms Act 1977 restricts how far liability can be excluded or limited — exclusions must be reasonable.
ABSOLUTE RULE	Liability for DEATH or PERSONAL INJURY caused by a defect CANNOT be limited or excluded by ANY contract clause. This is an absolute rule with no exceptions.
Consumer Sales (SGA 1979)	Sale of Goods Act 1979: goods sold to consumers must be fit for the purpose for which such goods are commonly supplied. Likely applies to retail/shrink-wrap software.
Bespoke Software (SGSA 1982)	Supply of Goods and Services Act 1982: only requires 'reasonable skill and care' for bespoke software. But UCTA 1977 still applies.
License Agreements	License may allow: one copy on one computer, server use up to agreed users, or site license for specified premises. Vendor protects own IP rights and limits customer usage.

Health & Safety

"It shall be the duty of every employer to ensure, so far as is reasonably practicable, the health, safety and welfare at work of all his employees." — Health and Safety at Work Act 1974.

- Provision and maintenance of safe plant and safe systems of work
- Such information, instruction, training and supervision as is necessary
- Safe working environment and adequate welfare arrangements
- Failure to comply with the act is a CRIMINAL offence
- The act also requires employers to ensure their activities do not risk the GENERAL PUBLIC's health and safety

The British Computer Society (BCS)

The BCS is the British national body for IT professionals. Other professional bodies include: IET (Institution of Engineering and Technology), CPSR (Computer Professionals for Social Responsibility), and ACM (Association for Computing Machinery — has its own Code of Ethics).

BCS Code of Conduct — 4 Pillars

1. Public Interest

- Regard for public health, safety and the environment
- Regard legitimate rights of third parties
- Have knowledge of, and comply with relevant legislation, regulations, standards
- No discrimination of any kind
- No active or passive bribery

2. Duty to Relevant Authority

- Apply due care and diligence
- If over-ruled — formally indicate the likely risks (CANNOT stay silent)
- Avoid conflicts of interest
- Do NOT disclose confidential information (with very limited exceptions)
- Do not withhold information on the performance of products

3. Duty to the Profession

- Uphold the reputation of the BCS
- Act with integrity at all times
- Consider possible consequences of statements on others
- Make public statements only if properly qualified to do so
- Notify BCS if convicted of a crime or becoming bankrupt

4. Professional Competence & Integrity

- Upgrade your professional knowledge and skills
- Do NOT claim competence that you do not possess
- Work ONLY within your professional competence
- Observe relevant clauses from the BCS Code of Good Practice
- Accept professional responsibility for your work

BCS Code of Good Practice

Practices Common to All Disciplines:

- Maintain your professional competence
- Follow regulations and applicable standards
- Act professionally in all dealings
- Use appropriate methods and tools
- Manage workload efficiently
- Participate maturely and show respect
- Promote good practice and represent the profession to the public

6 Key IT Practices:

1. Project Management	Managing related projects to improve organizational performance. Key factors: Governance, Alignment, Assurance, Management, Integration, Finances, Infrastructure, Planning, Improvement.
2. Relationship Management	Maintaining ongoing engagement with audience. Building client support and brand loyalty. Both B2C and B2B models.
3. Security	Protection of assets. Three main aspects: (1) Prevention, (2) Detection, (3) Re-action.
4. Safety Management	Preventing accidents and minimizing risk. Includes SMS documentation, emergency response planning, management commitment and responsibility, appointment of key safety personnel.
5. Change Management	Preparing and supporting individuals to adopt change. Three levels: Individual Change Management, Organizational/Initiative Change Management, and Enterprise Change Management Capability.
6. Quality Management	Overseeing all activities to maintain desired level of excellence. Also known as TQM (Total Quality Management). Achieves long-term goals through short-term initiatives.

Business Functions (Code of Good Practice):

- Requirements Analysis & Specification
- Software Development
- System Installation
- Training
- System Operations
- Support and Maintenance

EXAM STRATEGY: (1) Read scenario carefully. (2) Identify which lecture(s) apply.
(3) List specific laws and clauses. (4) Apply them to the SPECIFIC FACTS.
Always cite: exact law name + year + offence number + clause. That is where marks are earned.

Case Study 1 | L9a | Computer Misuse Act 1990

Scenario: The Disgruntled Employee

Ali, a former IT employee at TechCorp, was terminated last Friday. Before clearing his desk, he used his still-active credentials to log into the company server and deleted all the product design files. The following Monday, from home (credentials not yet revoked), he logged in again and emailed the entire client database to his personal Gmail. He tells police he 'just wanted to see if he could still log in.'

Questions:

1. Under CMA 1990, which specific offences did Ali commit?
2. What are the punishments for each offence?
3. Does it matter legally that his credentials were not yet revoked?

MODEL ANSWER:

Offence 3 — Unauthorized Alteration [Most Serious]	Deleting the product design files = altering and destroying computer data without permission. As an ex-employee he had NO authorization to alter company data. Punishment: Up to 5 years prison + unlimited fine.
Offence 2 — Access with Intent	Logging in from home AND emailing the client database to himself = unauthorized access with intent to commit a further offence (data theft). Punishment: Up to 5 years prison + unlimited fine.
Offence 1 — Unauthorized Access	Even the initial login 'just to see' counts as unauthorized access to computer material. Punishment: Up to 6 months prison + up to £5,000 fine.
KEY LEGAL POINT — The Credential Question	The fact that his credentials technically still worked does NOT make the access legally authorized. Authorization under CMA 1990 is LEGAL, not technical. Ali was no longer an employee; therefore all access was legally unauthorized regardless of whether the system allowed him in technically.

Case Study 2 | L9b | Pakistan Cyber Laws (All 3)

Scenario: The Cyber Cafe Fraudster

Bilal operates a cyber cafe in Lahore. He secretly installs keylogging software on all 20 computers. Over 3 months he collects banking credentials and passwords from customers. He transfers money from 15 victims' bank accounts and sends thousands of spam emails from stolen accounts advertising fake pharmaceutical products. When arrested by FIA, he claims: 'I just run a cafe. I'm not responsible for what software does on my machines.'

Questions:

1. Which Pakistan cyber laws apply to Bilal?

2. What specific charges can FIA bring under the 2007 Bill — with exact punishments?
3. What obligations does Bilal have as a cafe owner under the 2015 Act?

MODEL ANSWER:

Applicable Laws	Electronic Crimes Bill 2007 + PECA 2015 both apply.
Charges Under Bill 2007	Criminal Data Access: 3 years + Rs. 3 Lac (collecting credentials via keylogger). Electronic Fraud: 7 years + Rs. 7 Lac (stealing from bank accounts). Malicious Code: 5 years + Rs. 5 Lac (installing keylogging software). System Damage: 3 years + Rs. 3 Lac (accessing customers' accounts). Cyber Spamming: 6 months + Rs. 50,000 (mass spam emails). All charges are cumulative — Bilal faces very long total imprisonment.
FIA Authority	Under the 2007 Bill, FIA has EXCLUSIVE powers to investigate and charge these cases. No other police body can handle cybercrime under this law.
Obligations Under PECA 2015	Section 26 requires cyber cafes and all internet facility providers to HOLD DATA RECORDS for a minimum of 3 months. Bilal's own cafe logs are legally required evidence NOW being used against him.
Bilal's Defense Fails	The Bill applies to EVERY person who commits an offence, irrespective of nationality or citizenship. Deliberately installing keylogging software is an intentional criminal act — 'just running a cafe' provides zero legal shield.

Case Study 3 | L10 | Fixed Price Contract — Delays & Variation

Scenario: The Disputed Delay

DevSoft agreed to build a hospital management system for MedClinic under a fixed-price contract of Rs. 10 million. The contract includes a penalty clause: Rs. 50,000 per week delay, maximum Rs. 1 million. MedClinic changed requirements three times without following the contract's formal variation procedure, causing DevSoft to restart a key module. DevSoft is now 8 weeks late. MedClinic demands full penalty deductions (Rs. 400,000). DevSoft argues the delays were caused by MedClinic's unauthorized changes.

Questions:

1. Who is responsible for the delays — and under what contract clause?
2. Can MedClinic apply the full penalty clause here?
3. What specific contract provisions protect DevSoft?
4. What is the contractual significance of progress meeting minutes?

MODEL ANSWER:

Who is Responsible?	MedClinic bears significant responsibility. The 'Payments for Delays and Changes' clause states: when the CLIENT fails to meet its obligations (e.g. following the variation procedure), the developer is entitled to compensation for wasted effort and extra work. MedClinic bypassed the formal variation procedure three times — this is a CLIENT breach.
Can the Penalty Clause Apply?	The penalty clause (Rs. 50K/week, max Rs. 1M) is designed for delays CAUSED BY THE DEVELOPER. Since delays were triggered by MedClinic's unauthorized changes, the full penalty cannot fairly apply. DevSoft has strong grounds to challenge it and potentially counterclaim.

DevSoft's Protections	1. Variation Procedure: MedClinic bypassed it — extra work can still be claimed at agreed daily rates from the annex. 2. Payments for Delays & Changes clause: explicitly covers this situation. 3. Client Obligations clause: MedClinic had obligations they breached. 4. Daily Rate Annex: specifies charging rates per grade of staff for extra work.
Significance of Progress Meeting Minutes	'The minutes of meetings, duly approved and signed, should have contractual significance.' They constitute evidence of: milestones reached/missed, agreements made, and changes requested (by whom, when). If DevSoft documented each change request with cost and timeline impact in SIGNED minutes — this is legally binding evidence protecting them.

Case Study 4 | L10 | Liability for Defective Software

Scenario: The Faulty Medical Software

PharmaSoft sold drug dosage calculation software to a hospital under a standard license agreement. Their standard terms limited ALL liability to the purchase price (Rs. 200,000). A bug caused a patient to receive 10x the correct drug dose, resulting in serious injury. The hospital sues PharmaSoft for Rs. 50 million in damages. PharmaSoft points to the liability limitation clause.

Questions:

1. Is the liability limitation clause enforceable in this case?
2. What does the Unfair Contract Terms Act 1977 say about this?
3. Can PharmaSoft limit liability for death or personal injury?
4. Does Sale of Goods Act or Supply of Goods & Services Act apply?

MODEL ANSWER:

Is the Clause Enforceable?	PharmaSoft's clause limiting liability to Rs. 200,000 is legally valid for financial losses. However, for death or personal injury, it is NOT enforceable — this is an absolute rule.
Unfair Contract Terms Act 1977 (UCTA)	UCTA 1977 restricts how far liability can be excluded or limited. ABSOLUTE RULE: 'It is not possible to limit the damages payable if a defect in the product causes death or injury.' This rule cannot be overridden by any contract clause, no matter how clearly worded.
Result for PharmaSoft	Since the patient suffered serious personal injury, PharmaSoft CANNOT rely on the limitation clause for these damages. They are potentially liable for the FULL Rs. 50 million claim.
Goods vs. Services Analysis	Retail/shrink-wrap software: likely Sale of Goods Act 1979 — must be 'fit for purpose.' Bespoke software: Supply of Goods and Services Act 1982 — only 'reasonable skill and care.' Medical dosage software under license: 'fit for purpose' almost certainly applies — a dosage calculator giving wrong doses is clearly NOT fit for purpose. Even under the 1982 Act, UCTA 1977 prevents unreasonable exclusions.

Case Study 5 | L10 | Outsourcing Contracts & SLA

Scenario: The Outsourcing Disaster

The Government of Punjab outsourced its IT department to OutSource-IT for 5 years. The contract had no detailed Service Level Agreement. After 2 years: server uptime dropped to 60%, critical citizen data was permanently lost during a server migration, and OutSource-IT refused to allow government auditors to inspect their systems. The government wants to terminate the contract.

Questions:

1. What critical contract provisions were missing?
2. What is an SLA and why is it THE most critical element in IT outsourcing?
3. What should the audit rights provision have said?
4. What are the government's termination rights?

MODEL ANSWER:

Missing Critical Provisions	1. SLA — MISSING (most critical). 2. Performance Standards — not defined. 3. Audit Rights — MISSING (hence cannot be enforced). 4. Contingency Plan & Disaster Recovery — MISSING. 5. Data Ownership — not addressed.
What is an SLA and Why Critical?	An SLA (Service Level Agreement) is so important in IT outsourcing it is 'often treated as a separate agreement.' A proper SLA defines: uptime guarantees (e.g. 99.9%), response times, data backup frequency, escalation procedures, and consequences for SLA breaches. Without it, there is nothing measurable to enforce.
Audit Rights Provision	The contract must explicitly grant the government the right to inspect OutSource-IT's systems, records, and operations at any time. Example wording: 'The client reserves the right to audit the developer's systems, records, and operations at any time with 24 hours written notice.' Without this clause, refusal of auditors cannot be treated as a breach.
Termination Rights	The termination clause should have specified: payment for work done to date, compensation for staff redeployment, and immediate transfer of ALL citizen data and work products to the government. Without these provisions, termination becomes legally complex and expensive.

Case Study 6 | L11 | BCS Code of Conduct — All 4 Pillars

Scenario: The Overconfident BCS Member

Sara is an IT professional and BCS member hired as a cybersecurity consultant at a major bank. During the project she: (1) claims expertise in penetration testing she does not actually have, (2) shares the bank's internal security architecture 'casually over dinner' with a friend at a competing firm, (3) writes an anonymous but identifiable public blog post criticizing the bank's security, (4) when her manager instructs her to skip security testing to meet the deadline, she silently complies without objection.

Questions:

1. Which BCS Code of Conduct clause does each of Sara's 4 actions violate?
2. What should Sara have done in each situation?
3. What are the consequences for a BCS member who violates the Code?

MODEL ANSWER:

Violation 1 — False Competence Claim	Pillar violated: Professional Competence & Integrity — 'Do not claim competence that you do not possess. Work only within your professional competence.' Should have done: Disclosed limitation during hiring. Either declined the role or sought supervised assistance and disclosed this to the client.
Violation 2 — Disclosing Confidential Information	Pillar violated: Duty to Relevant Authority — 'Do not disclose confidential information' (with very limited legal exceptions — a dinner party is NOT one of them). Should have done: Maintained strict confidentiality. Never discussed client information outside of professional necessity, regardless of context.
Violation 3 — Unauthorized Public Statement	Pillar violated: Duty to Profession — 'Consider possible consequences of statements on others, make public statements only if properly qualified.' Being 'anonymous' does not make it acceptable if the client is identifiable. Should have never made the post.
Violation 4 — Silent Compliance	Pillar violated: Duty to Relevant Authority — 'Apply due care and diligence. If over-ruled, indicate the likely risks.' Being silent when risks exist IS ITSELF a violation. Should have written a formal risk memo documenting specific security risks, even if manager ultimately overruled her.
Consequences	BCS disciplinary proceedings can result in suspension or permanent removal of BCS membership. Separate legal liability (professional negligence, confidentiality breach) applies independently.

Case Study 7 | L10 | Contract Type Selection

Scenario: Wrong Contract for Vague Requirements

StartupXYZ wants to build an AI-powered recruitment platform but has only a rough idea of what features they need. DevTeam offers: Option A — Fixed Price at Rs. 20 million, or Option B — Time & Materials estimated at Rs. 12-18 million. StartupXYZ chooses Fixed Price because it 'seems safer.' During development, actual requirements are far more complex. DevTeam delivers 4 months late with Rs. 3 million in extra variation charges.

Questions:

1. Was Fixed Price the right contract choice given vague requirements?
2. Why would Time & Materials have been more appropriate?
3. How does a developer 'load' risk into a fixed price quote?
4. What should a proper Fixed Price contract have included?

MODEL ANSWER:

Was Fixed Price Right?	NO. Fixed price works well ONLY when requirements are clearly, fully, and precisely defined. StartupXYZ had only a 'rough idea.' The slides state: 'Often the work to be carried out is not sufficiently well specified for any developer to offer a fixed price.'
Why Time & Materials Was Better	1. 'Part of the developer's task will be to discover the requirements' — this is a core T&M; use case. 2. Requirements can evolve naturally without expensive formal variation procedures. 3. The Rs. 20M fixed price included a large contingency markup that T&M; avoids.
How Developers Load Fixed Price	'Developer always loads a fixed contract with a contingency allowance.' If development goes smoothly, the developer makes extra profit — this is their reward for accepting the risk. Vague requirements = large contingency = expensive fixed price.

What the Fixed Price Contract Should Have Had	Uniquely referenced, dated requirements specification. Formal variation procedure with cost calculation method. Daily rate annex for staff per grade. Clear acceptance testing criteria defined before development begins. Signed progress meeting minutes. Stage payment milestones tied to specific deliverables.
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Case Study 8 | ALL 4 LECTURES | Comprehensive — All 4 Lectures Combined

Scenario: The Pakistani Software Company and UK Bank

TechSolutions (Pakistan) is building a fraud detection system for a UK bank under a fixed-price contract. Senior developer Kamran (BCS member) discovers: (1) A colleague accesses the bank's live production servers 'just to explore' without authorization. (2) TechSolutions is using a proprietary third-party code library in the deliverable without a proper license. (3) Kamran's manager instructs him to not tell the bank about a critical security vulnerability discovered in the system, to avoid jeopardizing the delivery timeline.

Questions:

1. What CMA 1990 offences is the colleague committing — with exact punishments?
2. What contract liability issues arise from using unlicensed code?
3. What are Kamran's specific BCS Code of Conduct obligations?
4. What Pakistan laws (PECA 2015) could additionally apply?

MODEL ANSWER:

Issue 1 — CMA 1990 Offences (Lecture 9a)	Offence 1 (minimum): Unauthorized access — up to 6 months + £5,000 fine. Potentially Offence 2 (Access with Intent): If colleague viewed data, this escalates to 5 years + unlimited fine. Since this involves a UK bank's systems, UK CMA 1990 applies regardless of where the colleague is based.
Issue 2 — Contract Liability: Unlicensed Code (Lecture 10)	INDEMNITY CLAUSE: The contract should include an indemnity clause where 'each party indemnifies the other for liability arising from its own faults' regarding IP rights violations. TechSolutions caused this — they are liable to indemnify the bank. RIGHTS OWNERSHIP CLAUSE: Should state ALL components in the deliverable are properly licensed — TechSolutions is in breach. THIRD-PARTY LIABILITY: The code library owner can also sue TechSolutions for copyright infringement independently.
Issue 3 — Kamran's BCS Code Obligations (Lecture 11)	Public Interest: Must not allow a known security vulnerability to endanger the bank's customers. Duty to Relevant Authority: 'Apply due care and diligence. If over-ruled, indicate the likely risks.' Kamran MUST formally document the vulnerability in writing — silent compliance is a BCS violation. Professional Competence: 'Accept professional responsibility.' Cannot accept responsibility for a system with a known unfixed vulnerability. Required actions: Write formal email to management documenting vulnerability + risks; escalate colleague's server access; escalate IP licensing issue — all with written records.
Issue 4 — PECA 2015 (Lecture 9b)	If acts originate from Pakistan, PECA 2015 applies simultaneously with CMA 1990: Unauthorized access to bank's information system: 6 months imprisonment or Rs. 100,000 fine (or both). FIA has authority to investigate cases involving Pakistani nationals accessing foreign systems.

KEY EXAM INSIGHT

This scenario shows how all 4 lectures interconnect in real professional situations. A single situation simultaneously triggers: criminal law (CMA 1990 + PECA 2015), contract law (indemnity, rights ownership), and professional ethics (BCS Code of Conduct — all 4 pillars). The exam tests whether you identify ALL applicable frameworks, not just one.
